PLANE TABLE SURVEY

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- Generally, a graphic method
- Field work and plotting of map done simultaneously
- Suitable for filling in details between stations fixed by other methods (control points),
- Adopted for small scale mapping where great accuracy is not required.



EQUIPMENT

It mainly consists of the following equipment;

- **Drawing board mounted on tripod;**

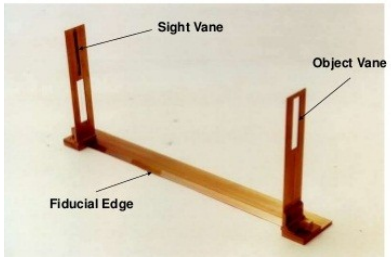
Made of well seasoned wood mounted on tripod can be rotated and clamped in any position can be leveled.



- **Alidade:**

Metal straight edge (ruler like), one edge is tapered and called fiducial edge. Can be plane or telescopic. Fitted with sight vanes. It has narrow slit on one side and central vertical wire on the other side.

ACCESSORIES



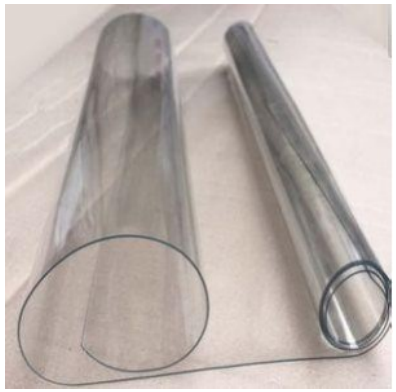
Alidade



Trough Compass



Plumb Bob



Water Proof Cover



Paper Sheet (Good Quality)

SETTING UP

- Convenient height
- Centering on control point
- Leveling of plane table

OREINTATION

- By magnetic compass
- By back sighting

METHODS

- Intersection
- Resection
- Radiation
- Traversing

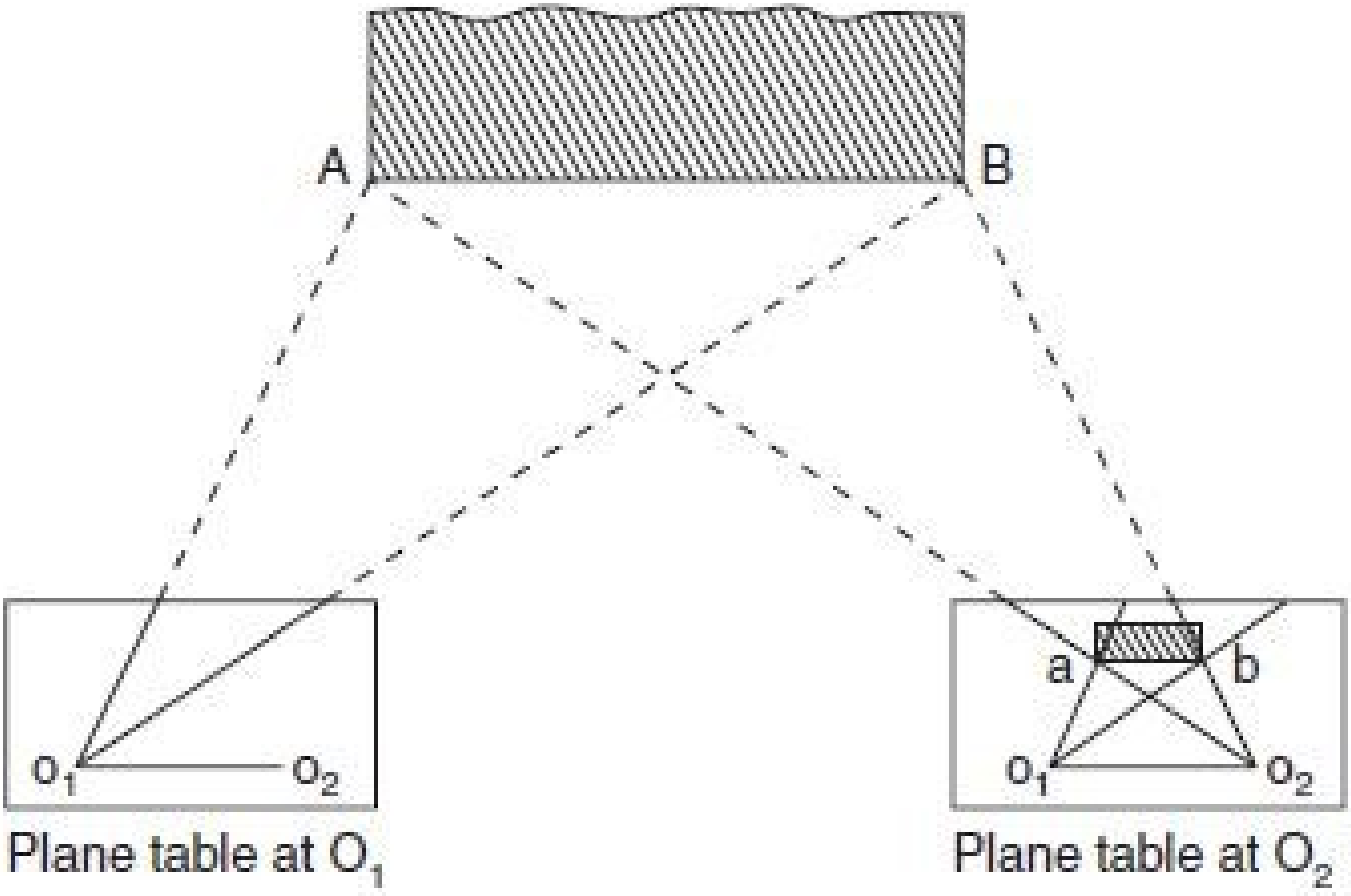


Fig. 14.8. Intersection method of plane tabling

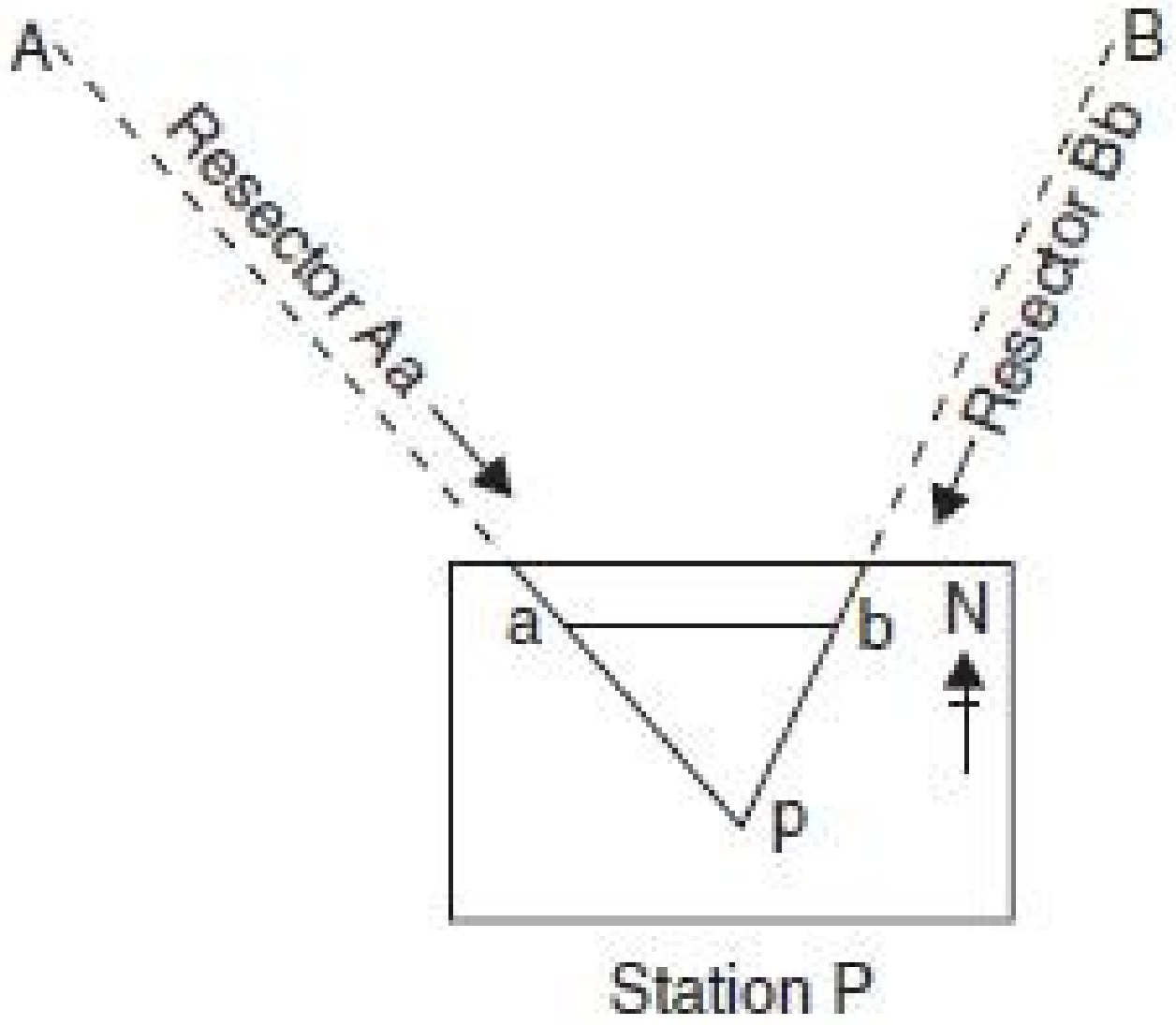


Fig. 14.10. Resection after orientation with compass

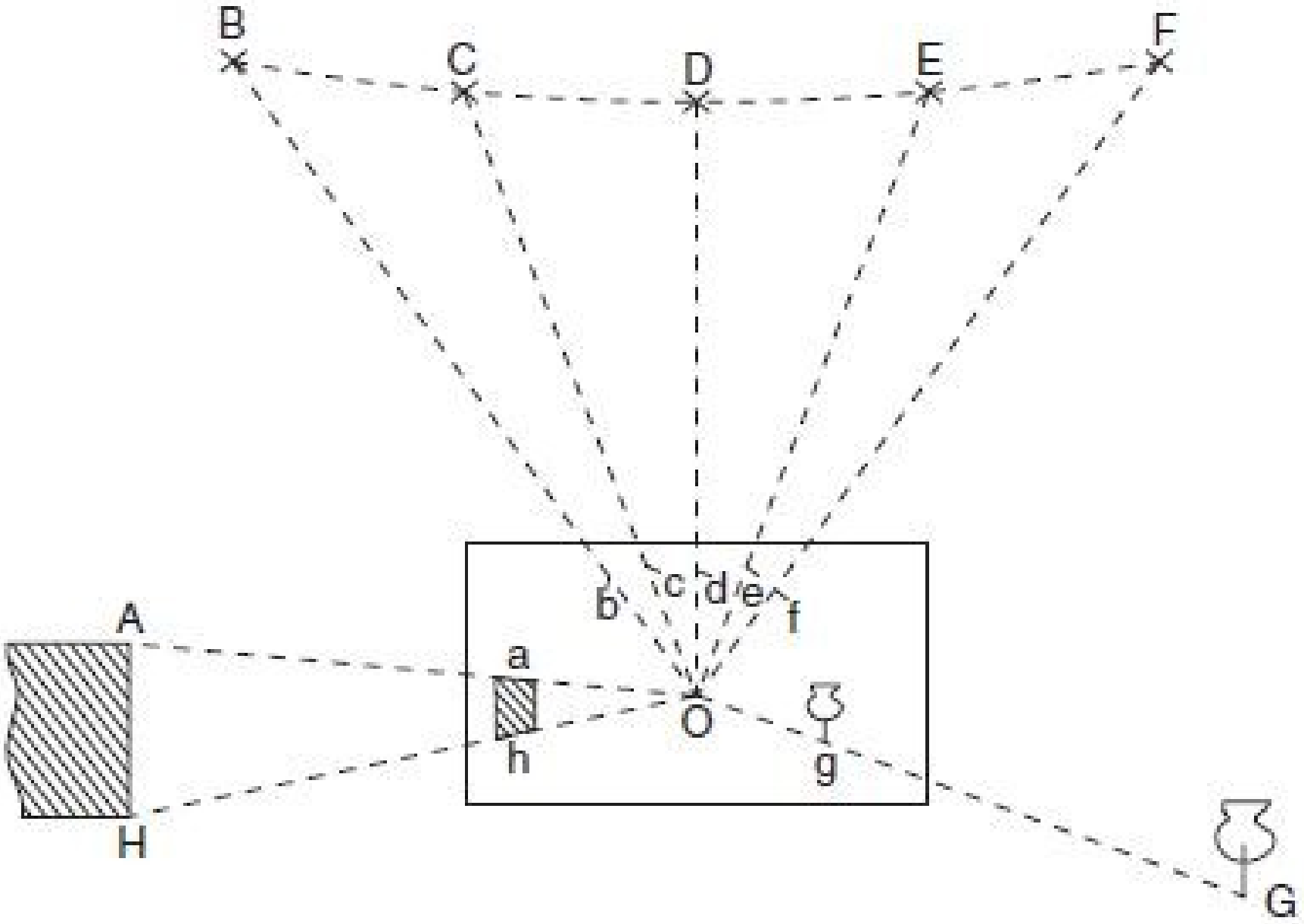


Fig. 14.7. Radiation method of plane tabling

ERRORS

- Board or Plane Table not horizontal or level.
- Board not centered properly on control point.
- Board not correctly oriented.
- Movement in table due to improper clamping.
- Inaccurate sighting.
- Alidade not correctly centered on point.
- Rays not drawn accurately.
- Inaccurate plotting.
- Expansion/ Contraction of paper.

ADVANTAGES

- Suitable for small scale mapping.
- Most rapid.
- Comparison with ground all the time for accuracy.
- Exact representation of ground.
- Omitting of details is not possible.
- Errors can be detected by check lines easily.
- Contours will represent profile of ground correctly.
- Great skills not required.

DISADVANTAGES

- Work only possible in fair whether.
- Equipment is heavy/cumbersome.
- Accessories are small, may be lost.
- Not very accurate for large scale survey.